

Institute of Business Administration – Mid-Term Exam

PART B: Subjective – 90 minutes / 35 marks

For official use only:

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Name: _____ ERP # _____

Course: CSE248 - Computer Communications & Networking	Exam: Mid Term
Instructors: Tasbiha Fatima / Waseem Arain	Total Time: 90 Minutes
Class: BSCS	Date: 19 th March, 2025
Instructions	
- Attempt all questions - Use only the space provided here to answer, so write carefully and legibly. - Understanding the question paper is part of the exam, so no explanations will be offered on questions.	

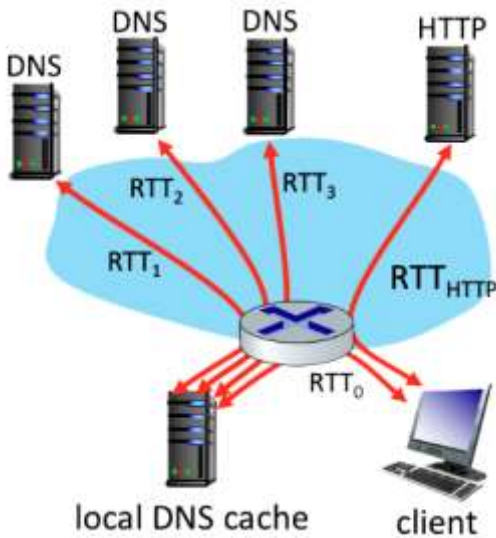
1. Consider a packet of length L that begins at end system A and travels over three links to a destination end system. These three links are connected by two packet switches. **(4)**

- a. Let d_i , s_i , and R_i denote the length, propagation speed, and the transmission rate of link i , for $i = 1, 2, 3$. The packet switch delays each packet by d_{proc} . Assuming no queuing delays, in terms of d_i , s_i , R_i , ($i = 1, 2, 3$), and L , what is the total end-to-end delay for the packet?

- b. Suppose now the packet is 1,500 bytes, the propagation speed on all three links is 2.5×10^8 m/s, the transmission rates of all three links are 2.5 Mbps, the packet switch processing delay is 3 msec, the length of the first link is 5,000 km, the length of the second link is 4,000 km, and the length of the last link is 1,000 km. For these values, what is the end-to-end delay?

- c. Now, suppose $R_1 = R_2 = R_3 = R$ and $d_{\text{proc}} = 0$. Further suppose that the packet switch does not store-and-forward packets but instead immediately transmits each bit it receives before waiting for the entire packet to arrive. What is the end-to-end delay?

2. Suppose within your Web browser you click on a link to obtain a Web page. The IP address for the associated URL is not cached in your local host, so a DNS lookup is necessary to obtain the IP address. Examine the topology and suppose that four DNS servers, as shown below, are visited before your host receives the IP address from DNS. The first DNS server visited is the local DNS cache, with an RTT delay of $RTT_0 = 5$ msec. The second, third and fourth DNS servers contacted have RTTs of 43, 24, and 31 msec, respectively. Initially, let's suppose that the Web page associated with the link contains exactly one object, consisting of a small amount of HTML text. Suppose the RTT between the local host and the Web server containing the object is $RTT_{HTTP} = 22$ msec. (5)



- a) Assuming zero transmission time for the HTML object, how much time (in msec) elapses from when the client clicks on the link until the client receives the object?

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- b) Now suppose the HTML object references 4 very small objects on the same server. Neglecting transmission times, how much time (in msec) elapses from when the client clicks on the link until the base object and all 4 additional objects are received from web server at the client, assuming non-persistent HTTP and no parallel TCP connections?

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- c) Suppose the HTML object references 4 very small objects on the same server, but assume that the client is configured to support a maximum of 5 parallel TCP connections, with non-persistent HTTP.

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- d) Suppose the HTML object references 4 very small objects on the same server, but assume that the client is configured to support a maximum of 5 parallel TCP connections, with persistent HTTP.

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- e) What's the fastest method we've explored: Non-persistent-serial, Non-persistent-parallel, or Persistent-parallel?
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3. Is it possible for an organization's Web server and mail server to have exactly the same alias for a hostname (for example, Google.com)? What would be the type for the RR that contains the hostname of the mail server? (3)

4. Suppose you would like to urgently deliver 50 terabytes data from Boston to Los Angeles. You have available a 200 Mbps dedicated link for data transfer. Would you prefer to transmit the data via this link or instead use FedEx overnight delivery? Explain. (4)

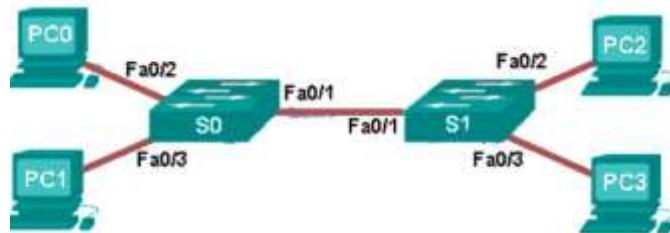
5. Following is the output of *show mac address-table* command on a switch (4)

```
S2# show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
All     0100.0ccc.cccc   STATIC    CPU
All     0100.0ccc.cccd   STATIC    CPU
...
All     0180.c200.000c   STATIC    CPU
All     0180.c200.000d   STATIC    CPU
All     0180.c200.000e   STATIC    CPU
All     0180.c200.000f   STATIC    CPU
All     0180.c200.0010   STATIC    CPU
All     ffff.ffff.ffff   STATIC    CPU
1       001a.e3cf.b883   DYNAMIC   Fa0/1
1       001a.e3cf.b8c0   DYNAMIC   Fa0/1
1       0050.56b3.27d6   DYNAMIC   Fa0/1
1       0050.56b3.ff54   DYNAMIC   Fa0/18
Total Mac Addresses for this criterion: 24
```

- a. What MAC addresses are recorded in the table? To which switch ports are they mapped and to which devices do they belong? Ignore MAC addresses that are mapped to the CPU.

b. Why do you see more than one MAC address mapped onto a single port above?

c. Refer to the exhibit. A ping to PC3 is issued from PC0, PC1, and PC2 in this exact order. Which MAC addresses will be contained in the S1 MAC address table that is associated with the Fa0/1 port?



6. In the generic SR protocol, the sender transmits a message as soon as it is available (if it is in the window) without waiting for an acknowledgment. Suppose now that we want an SR protocol that sends messages two at a time. That is, the sender will send a pair of messages and will send the next pair of messages only when it knows that both messages in the first pair have been received correctly. Suppose that the channel may lose messages but will not corrupt or reorder messages. Design an error-control protocol for the unidirectional reliable transfer of messages. Give an FSM description of the sender and receiver. Describe the format of the packets sent between sender and receiver, and vice versa. If you use any procedure calls other than `udt_send()`, `start_timer()`, `rdt_rcv()` etc. that we covered in the text, clearly state their actions. Give an example (a timeline trace of sender and receiver) showing how your protocol recovers from a lost packet.

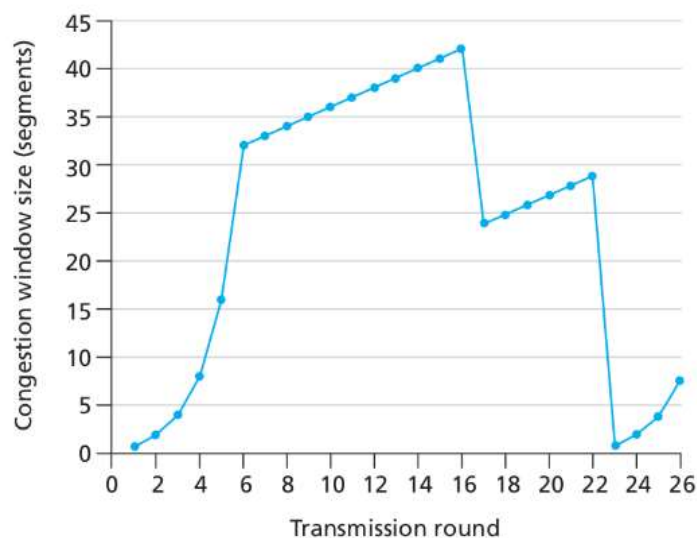
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7. Consider transferring an enormous file of L bytes from Host A to Host B. Assume an MSS of 536 bytes.
- What is the maximum value of L such that TCP sequence numbers are not exhausted? **(4)**

- For the L you obtain in (a), find how long it takes to transmit the file. Assume that a total of 66 bytes of transport, network, and data-link header are added to each segment before the resulting packet is sent out over a 155 Mbps link. Ignore flow control and congestion control so A can pump out the segments back to back and continuously.

8. Assuming TCP Reno is the protocol experiencing the behavior shown above, answer the following questions. In all cases, you should provide a short discussion justifying your answer.

(5)



- Identify the intervals of time when TCP slow start is operating.

- Identify the intervals of time when TCP congestion avoidance is operating.

- After the 16th transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?

- After the 22nd transmission round, is segment loss detected by a triple duplicate ACK or by a timeout?

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e. What is the initial value of ssthresh at the first transmission round?

f. What is the value of ssthresh at the 18th transmission round?

g. What is the value of ssthresh at the 24th transmission round?

h. Assuming a packet loss is detected after the 26th round by the receipt of a triple duplicate ACK, what will be the values of the congestion window size and of ssthresh?

i. Suppose TCP Tahoe is used (instead of TCP Reno), and assume that triple duplicate ACKs are received at the 16th round. What are the ssthresh and the congestion window size at the 19th round?
